Chapter 13

13.1.

Name	Type	Offset	Scope
operand1	int	0	main
operand2	int	-1	main
operation	char	-3	main
result	int	-2	main

13.3.

```
      if (a)
            x = b;
      else
            x = c;
```

13.5.

```
            AND         R0, R0, #0        ; init r0 at 0
            LDR         R1, R5, #0

            BRz         CASE_1            ; compare x==0
            ADD         R1, R1, #-1
            BRz         CASE_2            ; compare x==1
            BR          CASE_DEF    ; goto default case

      CASE_1:
            ADD         R1, R0, #3
            STR         R1, R5, #-1 ; y = 3

      CASE_2:
            ADD         R1, R0, #4
            STR         R1, R5, #-1 ; y = 4
            BR          END_SWITCH  ; break

      CASE_DEF:
            ADD         R1, R0, #5
            STR         R1, R5, #-1 ; y = 5
            BR          END_SWITCH  ; break

      END_SWITCH:
            .
            .
            .
```

13.7. This if-else statement **cannot** be converted into a switch statement. All cases labels must be integral constants. The if conditional (x == y) cannot be converted into a case label for the switch.

13.9. a. 0
b. 0
c. 11 4

13.11.

```
#include <stdio.h>
#define TRUE 1
#define FALSE 0

int main()
{
      char nextChar;   /* Next character in email address  */
      int gotAt  = FALSE;  /* Indicates if At @ was found  */
      int gotDot = FALSE;  /* Indicates if Dot . was found */
      int charCount = 0;

      printf("Enter your email address: ");

      do {
            scanf("%c", &nextChar);
            charCount++;

            if (nextChar == '@' && charCount > 1) {
               gotAt = TRUE;
               charCount = 0;
            }

            if (nextChar == '.' && gotAt == TRUE && charCount > 1) {
               gotDot = TRUE;
               charCount = 0;
            }
      }
      while (nextChar != ' ' && nextChar != '\n');

      if (gotAt == TRUE && gotDot == TRUE && charCount > 1)
         printf("Your email address appears to be valid.\n");
      else
         printf("Your email address is not valid!\n");
 }
```

13.13.

```
#include <stdio>

int main()
{
      int i;
      int sum;

      i = 0;

      do
      {
            if (i % 4 == 0)
                  sum = sum + 2;
            else if (i % 4 == 1)
                  sum = sum - 6;
            else if (i % 4 == 2)
                  sum = sum * 3;
            else if (i % 4 == 3)
                  sum = sum / 2;

            i++;

      }
      while (i <= 100);

      printf("%d\n", sum);
}
```

13.15.

a.
```
for ( ; condition; )
      loopbody;
```

b.
```
init;
while (condition)
{
      loopbody;
      reinit;
}
```

13.17. It counts the number of bits that are set in the two's complement representation of the integer provided by the user.

Questions in the text denoted by the question mark icon:

Page 353 It “echoes” the user input back to the monitor.

Page 355 Loop 1: 0 1 2 3 4 5 6 7 8 9 10
Loop 2: a b c d e f g h i j k l m n o p q r s t u v w x y z
Loop 3: Counts the number of bits that are set in inputValue